

Session 01 (AIML) - Assignment Questions

Q1. (Variables & Naming Rules)

Create variables to store the following information about yourself:

- Your full name (string)
- Your age (integer)
- Your height in meters (float)
- Whether you are a student (boolean)

Print all four variables. Then explain in comments why you chose those variable names and check their data types using `type()`.

Q2. (Data Types)

Write a Python program that:

- a) Creates one variable of each core data type: `str`, `int`, `float`, and `bool`.
 - b) Prints each variable along with its data type using `type()`.
 - c) Try converting the integer to float and float to integer. Print the results and explain the output.
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Q3. (Operators - Arithmetic)

Write a program that takes two numbers `a = 15` and `b = 4` and prints the results of:

- Addition, Subtraction, Multiplication, Division, Floor Division, Modulo, and Exponentiation.

Add comments explaining the difference between `/` and `//` operators with an example.

Q4. (Operators - Comparison & Logical)

Given:

```
x = 25
y = 30
```

```
z = 25
```

Write expressions using comparison and logical operators to check:

1. Is x greater than y?
2. Is x equal to z?
3. Is x less than or equal to y AND y is not equal to z?
4. Is x greater than y OR x equal to z?

Print all results with descriptive messages.

Q5. (Input/Output)

Write a program that asks the user to input:

- Their name
- Their birth year

Then calculate and print their current age (assume current year is 2026). Use proper formatting and add comments explaining why you need to convert the input.

Q6. (Combined - Variables, Operators & Input)

Create a simple **BMI Calculator**:

- Take height (in meters) and weight (in kg) as input from the user.
- Calculate BMI using the formula: $BMI = weight / (height ** 2)$
- Print the BMI rounded to 2 decimal places.

Include appropriate comments.

Q7. (Comments)

Write a small program that calculates the area and perimeter of a rectangle.

- Take length and width as input.
 - Use **at least 5 meaningful comments** (single-line and one multi-line comment).
 - Follow good commenting practices (explain *why*, not *what* for obvious lines).
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Q8. (Operators - Assignment & Membership)

```
fruits = ["mango", "apple", "banana", "cherry"]  
score = 50
```

Perform the following operations:

1. Add 25 to score using assignment operator (+=).
 2. Check if "apple" is in the fruits list.
 3. Check if "grape" is **not** in the list.
- Print all results with clear messages.

Q9. (Mini Project - All Concepts)

Write a **Student Profile Generator** program that:

- Takes input: name, age, city, and marks in 3 subjects.
- Calculates total marks and percentage.
- Prints a formatted profile with all information.
- Use meaningful variable names, proper comments, and different data types.

Q10. (Debugging & Understanding)

The following code has some issues. Correct it and add proper comments explaining the fixes:

```
Name = "Alice"  
age 20  
city = Amsterdam  
  
print("My name is" + Name)  
print("I am" age "years old")  
print("I live in " city)  
  
# Check if age is greater than 18 and print message  
print("Adult:" age > 18)
```

